

REVIEWER'S REPORT
(Administrative Confidential)

Please review the enclosed paper and return the annotated manuscript and this review form by the date specified in the email.

Author(s): Adam Zdunek and Waldemar Rachowicz

Paper Title: Efficient jet-engine inlet RCS prediction using higher order FEM and reduced order modeling

1. Are the contents of this paper suitable for publication in the *Electromagnetics*? Yes
2. Does the paper contain new idea or result that has not been published previously? Yes
3. Are the contents scientifically correct? Yes
4. Are the ideas or results presented herein important for Electromagnetic Community? Yes

Confidential Comments to Editor (use as much space as you need, the review form may extend to the next page or pages)

The Authors present the computation of the RCS of an arbitrarily shaped electrically large scatterer by the use of hp-FEM. A domain decomposition method with high order guided modes transmission conditions is employed to reduce the computational memory requirements. Several tests are performed to show the accuracy improvements of the hp-FEM compared to the conventional h-refined or p-enriched finite elements.

As discussed by the Authors in section 4, it should be interesting to have more insight on the effect of the reduced system matrix sparsity with p-enrichments to the overall solution efficiency, the times required by the employed sparse direct solver.

Comments for Transmittal to the Authors (use as much space as you need)

The meaning of the “+” sign postposed to the operator K_1 in the LHS of equation (20a) is not clear.

At p. 21, the number of degrees of freedom (dof:s) is usually abbreviated as “DOFs” or “dofs”.

At p. 32, the sentence “The described bottleneck can alleviated by...” should read “The described bottleneck can be alleviated by...”.

Recommendation

_____Accept as it is

___X___Acceptable after minor revisions (see comments)

_____Acceptable after major revisions (see comments)

_____Reject

___01/09/2013___
Date

___Laurent Ntibarikure___
Reviewer's name

Laurent Ntibarikure
Reviewer's signature